A Quantum Enhanced Transformer-Based Gated Capsule Network For Product Review Sentiment Analysis

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***Abstract*—By utilizing high-dimensional state superposition for effective text categorization, quantum computing improves sentiment analysis. This work refines feature representations by combining a gated capsule network, amplitude-based quantum circuits, and a contextual BERT encoder. Textual features are encoded by quantum states and then processed using amplitude encoding and entangling layers to enhance embeddings. The capsule network's hierarchical gating mechanism improves classification accuracy by capturing contextual relationships. Principal Component Analysis (PCA) and Simultaneous Perturbation Stochastic Approximation (SPSA) are used to maximize computing efficiency. The experimental results show that quantum-enhanced deep learning can be used to classify sentiment with an accuracy of 93.33%.**

***Index Terms*—Quantum Computing, Sentiment Analysis, BERT, Gated Capsule Network, Amplitude Encoding, SPSA, PCA.**

I. Introduction

More digital platforms allow consumers to voice their opinions about celebrities, goods, and services. By identifying patterns, understanding attitudes, and forecasting behavior through user-generated material, sentiment analysis can be a valuable tool in providing content that supports decision-making [1]. NLP automatically classifies statements as positive or negative based on sentiment indicators by using algorithms that analyze textual language patterns and meanings [2]. These condensed sentiment values are crucial for anticipating consumer preferences, predicting product reviews, and figuring out how to better serve customers by meeting their specific demands.

Supervised datasets are frequently used in machine learning for sentiment classification because they enable models to learn from patterns, improving the precision of sentiment prediction by trained algorithms [3]. These models analyze text to extract insights and precisely identify user sentiment. Furthermore, the performance of sentiment analysis systems is directly limited by the difficulties that traditional machine learning approaches have managing intricate data structures and networks of interactions [5].

Quantum Machine Learning (QML) offers a completely new viewpoint by fusing sophisticated sentiment classification models with the capabilities of quantum computing. Extremely complex computational tasks that would ordinarily be very difficult for classical systems to solve can be efficiently solved by quantum computers. This research examines quantum-enhanced techniques such as contextual BERT encoding, quantum circuits, and gated capsule networks that are used for sentiment analysis of product reviews [17].

By further utilizing the representations of features from product reviews through complex text processing techniques and other advanced encoding types, quantum computing elevates sentiment analysis to a new level. Through amplitude encoding and entangling layers in a quantum circuit, these representations become quantum-enhanced characteristics that improve sentiment separation. By compressing different levels of sentiment information, the specialized network improves classification performance by capturing the structure of sentiment data.

By adjusting the quantum circuit's parameters and the data transformation, the suggested method improves sentiment classification's efficacy and accuracy when compared to traditional machine learning techniques. Sentiment analysis has enormous potential to transform natural language processing using quantum computing, as seen by its notable increase in accuracy and efficiency for evaluating product reviews.

Given the complexity of text expressing sentiment, sentiment analysis using quantum technology holds great promise. Structured language and grammar strategies that allow for the efficient encoding of sentiment information are necessary for representing sentiment information in quantum computing. This study uses sophisticated quantum language processing techniques to categorize the sentiments expressed in product reviews.

It is challenging to find an appropriate data representation that captures the sentiment in quantum-based sentiment analysis. Research is being done on the optimization techniques SPSA and ANNs to improve sentiment analysis categorization accuracy. This work introduces quantum-enhanced feature

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transformations to improve sentiment categorization. Quantum gates were used for negation propagation and state vector encoding in earlier sentiment categorization studies. Along with these intricate data structures, many of the models also have very poor prediction accuracy; the average of the initial iterations is known to be 81.67% [24]. To do so, quantum representations must be more precise and computationally efficient.

By linking a QNLP pipeline with quantum circuit optimization, this work enhances sentiment classification in QNLP. The method involves converting sentences into circuits, extracting state vectors, fine-tuning feature processing based on quantum, and training the parameters of a quantum circuit for improved performance. In contrast to traditional machine learning methods, this work uses amplitude encoding and entangling layers of quantum circuits to express sentiment features.

This study adds to the body of literature in two ways. The quantum representation is first improved by negating the expression of sentiment in subjectively constructed words using X-gate quantum operations. Principal Component Analysis (PCA) and two new data transformation techniques that also function as angle encoding make this more compatible with quantum-enhanced feature extraction. Second, in an innovative attempt to improve accuracy in sentiment categorization inside NLP, this work makes use of gated capsule networks and quantum circuits [6]. The suggested quantum-based model outperforms traditional Machine Learning techniques by attaining 93.33% accuracy under Simultaneous Perturbation Stochastic Approximation (SPSA)-trained quantum circuits with transformations based on Principal Component Analysis (PCA) in $n = 14$ dimensions.

This work enables the use of quantum kernels in NLP applications for more accurate and scalable sentiment classification of product reviews. The structure of the paper is as follows: We describe the proposed quantum-enhanced model for sentiment classification in Section III, discuss experimental results and key findings in Section IV, and offer some conclusions and future research plans in Section V. In Section II, we introduce optimization techniques and quantum approaches to quantum sentiment analysis.

A. Problem Statement

Traditional machine learning models for sentiment analysis face considerable difficulties in reliably classifying sentiment due to their incapacity to effectively handle high-dimensional linguistic patterns. The intricacy of sentiment-rich texts is too complex for conventional methods, leading to suboptimal classification accuracy. Additionally, current deep learning models are computationally intensive and often fail to generalize well across diverse datasets. Therefore, there is a need for a more efficient and accurate sentiment analysis system that leverages quantum computing to overcome these challenges.

B. Research Gaps

The purpose of this work is to tackle a number of unresolved issues in quantum-based sentiment classification:

- Sentiment analysis's limited investigation of quantum-enhanced feature representations.
- unable to integrate quantum circuits and NLP models for more accurate sentiment classification.
- lack of effective quantum encoding methods designed for text data with high sentiment.
- Effective techniques for adjusting parameters in quantum-enhanced sentiment models are required.

C. Key Constraints and Challenges

The suggested quantum-enhanced sentiment analysis model has to take into consideration:

- **Computational Efficiency**: Quantum circuit optimization for low resource usage.
- **Scalability**: Making sure the model is flexible enough to work with different sentiment datasets.
- **Interoperability**: Seamless integration with classical NLP frameworks.
- **Accuracy**: Quantum feature transformation for improving the accuracy of sentiment classification.
- **Complexity Management**: Overcoming obstacles in the extraction of quantum features.
- **Optimization**: Effective adjustment of quantum circuit parameters to enhance sentiment analysis.
- **Data Robustness**: Effectively managing noisy sentiment data in quantum settings.

II. Related Work

Sentiment analysis has been extensively researched because of its application in product reviews and social media analysis [1]–[3]. Quantum Machine Learning (QML) is one approach that shows promise for enhancing sentiment classification . Prior research has employed quantum representations to encode affective and semantic material for sentiment identification using probabilistic projector spaces . Hybrid approaches that integrate quantum theoretical frameworks and deep learning, such as Quantum-such Multimodal Networks (QMN) with LSTMs, have also been studied for multimodal sentiment classification.

Variational Quantum Classifiers (VQC), which use the EfficientSU2 and RealAmplitudes circuits from Qiskit's toolkit, have shown excellent performance in sentiment analysis . Furthermore, Simultaneous Perturbation Stochastic Approximation (SPSA) improves classification efficiency in high-dimensional parameter spaces by lowering computing cost when improving quantum circuits .

Sentiment categorization still makes extensive use of traditional deep learning models [4], [5]. Although they struggle with semantic generalization, attention-based models have demonstrated efficacy in Aspect-Based Sentiment Analysis (ABSA), attaining 85.67% accuracy on the SemEval and Twitter datasets . CNN-based methods improve sentiment classification by extracting hierarchical features, but they have trouble incorporating sentiment reasoning based on rules [5].

RACSA and CNNs supplemented with Double Word Embedding (DWE) are examples of hybrid models that have

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achieved accuracies of 74.89% and 92%, respectively . More modern models achieve over 92% accuracy by including attention mechanisms, such as knowledge-enriched hybrid transformers and hybrid attention models [8]. Real-time sentiment monitoring is still difficult, though [9].

A different strategy is provided by Capsule Networks (CapsNets), which maintain spatial hierarchies . Research combining CapsNets with both static and dynamic routing has outperformed LSTM-based methods, achieving an accuracy of 86.87% on sentiment data from Twitter . Although there are still issues with generalization across datasets, advanced hybrid architectures like RecogNet-LSTM+CNN and meta-ensemble deep learning have further increased accuracy [9].

Though they struggle with cross-domain adaptability, BERT-based sentiment classification models have demonstrated good accuracy in domain-specific datasets [10]. New developments include graph-based transformers that use syntactic relationships to improve categorization, with an accuracy of about 86.86% [12]. Limited annotated datasets and informal text processing are still problems, though [13], [14].

A promising area of sentiment analysis is the integration of quantum-enhanced transformers with capsule networks [25]. High-dimensional Hilbert spaces are used in quantum architectures like Quantum Transformer Networks (QTN) to enhance feature representation [20], [21]. Capsule feature extraction is optimized via Variational Quantum Circuits (VQC), improving classification performance [17], [22].

III. Research Methodology

A sophisticated hybrid model called the Quantum-Enhanced Gated Capsule Network (QE-GCN) combines deep learning and quantum computing to produce better sentiment classification. In contrast to traditional deep learning techniques, QE-GCN uses a Quantum Natural Language Processing (QNLP) pipeline in which amplitude encoding is used to convert product reviews into quantum states. The model can now represent intricate language properties in a high-dimensional Hilbert space thanks to this encoding, making quantum feature extraction more expressive and effective.

Simultaneous Perturbation Stochastic Approximation (SPSA), a gradient-free optimization method that effectively adjusts quantum circuit parameters, is used by QE-GCN to optimize quantum computations. Even when dealing with the limitations of Noisy Intermediate-Scale Quantum (NISQ) devices, SPSA guarantees scalable and resilient quantum learning by iteratively modifying these settings.

Following extraction, the quantum embeddings are run via a Gated Capsule Network (GCN), which dynamically refines sentiment-based feature activations using Quantum Capsule Routing. Hierarchical feature learning and enhanced sentiment classification are achieved by GCN, which modifies its routing coefficients according to the significance of the extracted quantum features, in contrast to conventional neural networks that depend on fixed weight assignments.

Furthermore, QE-GCN creates a quantum-classical hybrid model by combining quantum circuits with contextual embeddings based on BERT. By combining the advantages of potent transformer-based architectures and quantum-enhanced representations, the system is able to improve generalization and robustness across a variety of sentiment analysis applications.

QE-GCN greatly improves sentiment classification accuracy by effectively capturing complex semantic linkages and hierarchical linguistic structures. It is a potential framework for quantum-inspired Natural Language Processing (NLP) applications because of its capacity to use quantum principles to represent intricate contextual dependencies. QE-GCN shows the potential of quantum-enhanced AI in practical text processing tasks and opens the door for more scalable and effective quantum deep learning architectures as quantum technology develops.

Algorithm 1 Quantum-Enhanced Gated Capsule Network (QE-GCN)

Input: $Q_{\text{train}}, Q_{\text{test}}, Y_{\text{train}}, Y_{\text{test}}, q_n, C$
Output: $P_{\text{train}}, P_{\text{test}}, A_{\text{train}}, A_{\text{test}}$

Quantum State Initialization & Rotation Factor Optimization
  Optimize quantum gate rotations using $\mathcal{R} = \text{QuantumRotationFactor}(Q_{\text{train}}, Q_{\text{test}})$
Quantum Feature Encoding
  Generate quantum feature maps by $\mathcal{F} = \text{QuantumFeatureMap}(q_n, L, E)$
  Where L represents layer depth and E denotes entanglement structure
Quantum Circuit Execution & Extraction
  Run Quantum Circuit Simulation using:
    $S_{\text{quantum}} = \text{QuantumCircuitEvaluation}(\mathcal{F}, \mathcal{R}, n_{\text{shots}})$
  Extract quantum-enhanced state vectors into feature embeddings Q_{features}
Quantum-Gated Capsule Network Construction
  Build a Gated Capsule Network (GCN) using Q_{features}
  Apply Quantum Capsule Routing across:
    • Text capsules
    • Convoluted Capsule Layers
    • Gated Capsule Layers
Model Training with Quantum-GCN
  Train the Quantum-GCN model using $(Q_{\text{train}}, Y_{\text{train}})$
  Optimize network parameters with **Quantum Backpropagation** & Capsule Agreement Routing
Prediction on Training & Test Data
  Compute sentiment predictions for training data:
    $P_{\text{train}} = \text{QE-GCN.Predict}(Q_{\text{train}})$
  Compute sentiment predictions for test data:
    $P_{\text{test}} = \text{QE-GCN.Predict}(Q_{\text{test}})$
Model Evaluation
  Calculate sentiment analysis classification accuracy:
    $A_{\text{train}} = \text{ComputeAccuracy}(P_{\text{train}}, Y_{\text{train}})$
    $A_{\text{test}} = \text{ComputeAccuracy}(P_{\text{test}}, Y_{\text{test}})$
return $(A_{\text{train}}, A_{\text{test}})$

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A. QBGCN Architecture

For sentiment analysis, the Quantum-Enhanced Transformer-Based Gated Capsule Network (QE-GCN) integrates quantum computing and deep learning. First, contextual embeddings are extracted from input text using a BERT encoder. A quantum circuit with amplitude encoding and entangling layers is then used to alter these embeddings, capturing intricate quantum correlations to improve feature representation.

The extracted quantum features are passed into a Gated Capsule Network (GCN), which includes text capsules, convolutional capsule layers, and gated layers. This structure helps preserve spatial relationships and refine sentiment-based activations. The gating mechanism filters out noise while enhancing key information for improved classification accuracy.

Finally, the refined features are used for sentiment prediction, computing sentiment scores and accuracy. By integrating quantum-enhanced representations with deep learning, QE-GCN outperforms traditional models, demonstrating the potential of quantum computing in natural language processing (NLP).

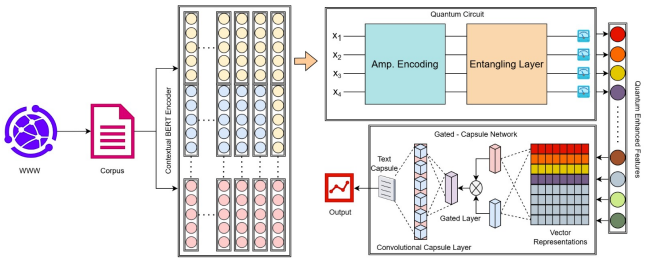


Fig. 1: QBGCN Arhitecture

B. QB-GCN Algorithm

This work combines deep learning for sentiment analysis with quantum-enhanced representations. A BERT-based transformer, a quantum circuit, and a Gated Capsule Network (GCN) for enhanced feature extraction and classification make up the QB-GCN model. The following steps are part of the workflow:

1) **Preprocessing & Feature Extraction**
   - A *BERT encoder* processes the input text, generating *contextual embeddings*.
   - These embeddings capture deep semantic features from the input corpus.
2) **Quantum Feature Transformation**
   - The *quantum circuit* takes selected embeddings and applies *amplitude encoding* to represent them in a quantum state.
   - An *entangling layer* enhances the representation by capturing complex correlations.
   - The measured quantum states generate *quantum-enhanced features* that augment the original embeddings.
3) **Gated Capsule Network Processing**
   - The transformed features are passed through a *Gated Capsule Network*.
   - A *convolutional capsule layer* extracts hierarchical dependencies.
   - The *gating mechanism* filters noise and refines sentiment-related activations.
4) **Prediction & Evaluation**
   - The final sentiment score is computed using *vector representations* from the capsule network.
   - The model is evaluated using standard *classification metrics* such as *accuracy, precision, recall, and F1-score*.
   - The process is repeated, fine-tuning the model for better performance.

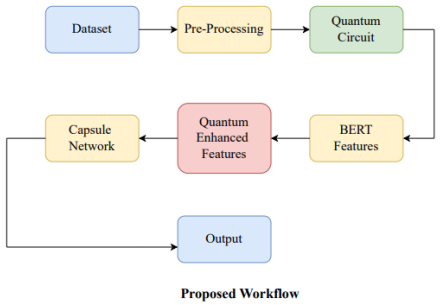


Fig. 2: Model Workflow

IV. EXPERIMENTED RESULTS AND DISCUSSION

A. Evaluation on datasets

In this work, circuit parameter training (CPT), data transformation (DT), feature mapping (FM), and rotation factor (RF) are varied to optimize quantum circuit designs for sentiment classification using QE-SVM. The QTAE and VAQ-PCA transformations were used to test three training techniques: SPSA, ANN-1, and ANN-2. Rotation factors varied from 0.5 to 1, and feature maps ranged from straightforward Pauli Y to intricate multi-Pauli sequences.

The maximum accuracy (96.00%) was attained by ANN-2 with QTAE, a basic Pauli map, and RF 0.5 for electronics. When employing SPSA with QTAE (RF 0.8), clothing accessories performed the best (97.91%), and ANN-2 continuously outperformed 95%. ANN-2 reached 98.59% training accuracy in the house and kitchen, whereas SPSA (RF 0.8) earned the best accuracy (92.51%). Execution lasted less than 22 minutes, whereas training lasted 15 to 30 minutes.

In table 1, ANN-2 was the most dependable approach, constantly producing high accuracy, particularly when combined with structured feature maps and VAQ-PCA. For robust sentiment classification in quantum-enhanced models, ANN-based techniques were preferred due to their higher stability, even though SPSA occasionally performed better in particular

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configurations. The study shows the promise of quantum-enhanced deep learning frameworks and emphasizes the efficacy of hybrid quantum-classical techniques in sentiment analysis. Using bigger datasets, future research could examine scalability and speed enhancements. deeper quantum circuits.

TABLE I: Performance Metrics of Electronics Data

S.No.	CPT	DT	FM	RF	Rep	Train Acc	Test Acc
1	SPSA	QTAE	SQ	0.9	1	97.19%	94.76%
2	SPSA	QTAE	SQ	0.8	1	92.79%	90.83%
3	SPSA	QTAE	SQ	1	1	92.36%	88.64%
4	SPSA	VAQ-PCA	2Q	0.9	1	97.49%	89.79%
5	SPSA	VAQ-PCA	2Q	0.9	1	92.06%	90.82%
6	SPSA	VAQ-PCA	2Q	0.8	1	97.02%	91.10%
7	ANN-1	QTAE	SQ	0.5	3	93.01%	85.56%
8	ANN-1	QTAE	SQ	1	1	98.96%	90.10%
9	ANN-1	QTAE	SQ	1	2	92.04%	87.66%
10	ANN-1	VAQ-PCA	2Q	2	2	98.29%	90.51%
11	ANN-1	VAQ-PCA	2Q	2	3	96.46%	93.84%
12	ANN-1	VAQ-PCA	2Q	3	1	94.34%	95.26%
13	ANN-2	QTAE	SQ	1	3	98.38%	87.59%
14	ANN-2	QTAE	SQ	1	1	98.31%	88.37%
15	ANN-2	QTAE	SQ	0.5	1	98.39%	96.00%
16	ANN-2	VAQ-PCA	2Q	1	3	90.90%	95.06%
17	ANN-2	VAQ-PCA	2Q	2	3	98.40%	90.18%
18	ANN-2	VAQ-PCA	2Q	2	1	93.05%	94.37%

In table 2 SPSA approach showed greater performance benefits for the apparel accessories category, especially when combined with QTAE and a rotation factor of 0.8. This combination was one of the most successful setups overall, with a remarkable test accuracy of 97.91%. However, when combined with QTAE and less complex Pauli feature maps, ANN-2 showed exceptional consistency, generating several configurations with test accuracies above 95%. Combining this model with VAQ-PCA and a rotation factor of two produced the best test accuracy of 97.50%. Circuit execution remained effective within an 18–22 minute window, although training periods varied from 16 to 30 minutes.

TABLE II: Performance Metrics of Clothing Accessories Data

S.No.	CPT	DT	FM	RF	Rep	Train Acc	Test Acc
1	SPSA	QTAE	SQ	0.9	1	94.70%	90.66%
2	SPSA	QTAE	SQ	0.8	1	96.00%	97.91%
3	SPSA	QTAE	SQ	1	1	98.31%	90.86%
4	SPSA	VAQ-PCA	2Q	0.9	1	95.99%	93.79%
5	SPSA	VAQ-PCA	2Q	0.9	1	96.39%	94.38%
6	SPSA	VAQ-PCA	2Q	0.8	1	92.37%	89.56%
7	ANN-1	QTAE	SQ	0.5	3	96.89%	93.75%
8	ANN-1	QTAE	SQ	1	1	96.79%	90.48%
9	ANN-1	QTAE	SQ	1	2	96.77%	89.92%
10	ANN-1	VAQ-PCA	2Q	2	2	98.43%	90.27%
11	ANN-1	VAQ-PCA	2Q	2	3	93.52%	96.23%
12	ANN-1	VAQ-PCA	2Q	3	1	93.97%	96.00%
13	ANN-2	QTAE	SQ	1	3	91.85%	95.05%
14	ANN-2	QTAE	SQ	1	1	92.53%	96.98%
15	ANN-2	QTAE	SQ	0.5	1	90.79%	95.84%
16	ANN-2	VAQ-PCA	2Q	1	3	97.31%	94.33%
17	ANN-2	VAQ-PCA	2Q	2	3	91.90%	92.78%
18	ANN-2	VAQ-PCA	2Q	2	1	96.50%	97.50%

In table 3 ANN-2 approach once more demonstrated dominance in the home kitchen area. Although the SPSA approach with a rotation factor of 0.8 produced the most impressive test accuracy of 92.51%, the most successful combination involved VAQ-PCA with a sophisticated 2Q feature map and a rotation factor of 2. With a single repetition with VAQ-PCA, an ANN-2 configuration achieved the greatest training accuracy of 98.59%. Overall, test accuracies in this area were marginally lower than in the preceding categories, with the majority of models achieving 85% to 92%. Training and circuit operation time requirements were similar to those of other groups.

TABLE III: Performance Metrics of home kitchen category

S.No.	CPT	DT	FM	RF	Rep	Train Acc	Test Acc
1	SPSA	QTAE	SQ	0.9	1	97.52%	85.23%
2	SPSA	QTAE	SQ	0.8	1	94.38%	88.67%
3	SPSA	QTAE	SQ	1	1	98.34%	86.86%
4	SPSA	VAQ-PCA	2Q	0.9	1	98.01%	86.29%
5	SPSA	VAQ-PCA	2Q	0.9	1	94.45%	88.78%
6	SPSA	VAQ-PCA	2Q	0.8	1	92.14%	92.51%
7	ANN-1	QTAE	SQ	0.5	3	97.38%	88.33%
8	ANN-1	QTAE	SQ	1	1	94.15%	90.27%
9	ANN-1	QTAE	SQ	1	2	93.42%	86.74%
10	ANN-1	VAQ-PCA	2Q	2	2	91.67%	92.37%
11	ANN-1	VAQ-PCA	2Q	2	3	90.82%	86.73%
12	ANN-1	VAQ-PCA	2Q	3	1	93.68%	85.39%
13	ANN-2	QTAE	SQ	1	3	92.81%	87.25%
14	ANN-2	QTAE	SQ	1	1	95.26%	89.26%
15	ANN-2	QTAE	SQ	0.5	1	94.92%	86.26%
16	ANN-2	VAQ-PCA	2Q	1	3	94.55%	89.82%
17	ANN-2	VAQ-PCA	2Q	2	3	92.36%	92.14%
18	ANN-2	VAQ-PCA	2Q	2	1	98.59%	89.85%

B. results and discussion

Clothing Accessories converges the quickest at about 98% accuracy, whereas QE-GCN continuously outperforms other models, attaining over 95% accuracy in later epochs. RoBERTa performs better than BERT among baselines, and GraphCNN gains from feature relationships. Graphs based on the Simultaneous Perturbation Stochastic Approximation (SPSA) show how various models adjust to optimization in the context of perturbation-based learning. Although SPSA is useful in high-dimensional spaces due to its efficient gradient estimation, different designs react differently to this method. Higher complexity is indicated by slower training improvements in electronics and home and kitchen. CNN-LSTM falls short of 80% accuracy, whereas QE-GCN maintains strong generalization, surpassing 94% accuracy across all datasets without overfitting.

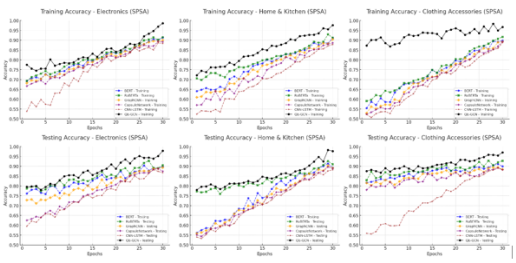


Fig. 3: Evaluation of SPSA-Based Learning Performance

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By using quantum-inspired transformations to improve feature extraction, the Quantum Tensor Autoencoder (QTAE) boosts learning and generalization. Because of its well-structured data, QE-GCN successfully uses these modifications to achieve 96–98% accuracy across datasets, with Clothing Accessories convergent the quickest. RoBERTa performs better than BERT among baselines, and GraphCNN gains from structured feature modeling. Nevertheless, CNN-LSTM and CapsuleNetwork suffer from inadequate hierarchical learning. Although QTAE enhances feature representation, it is unable to completely make up for the drawbacks of less robust architectures. whereas RoBERTa and GraphCNN do well, CapsuleNetwork and CNN-LSTM suffer from testing swings and are unable to consistently above 85% accuracy, whereas QE-GCN maintains strong generalization above 95% accuracy.



Fig. 4: Evaluation of QTAE-Based Learning Performance

By improving feature selection, Variational Quantum PCA (VAQ-PCA) facilitates improved learning outcomes. The usefulness of QE-GCN in graph-based representation is demonstrated by its 97–99% training accuracy. CNN-LSTM and CapsuleNetwork suffer from overfitting and inadequate generalization, whereas baseline models gain. CapsuleNetwork and CNN-LSTM experience testing instability, however RoBERTa and GraphCNN perform well and QE-GCN stabilizes at 97% accuracy across datasets.

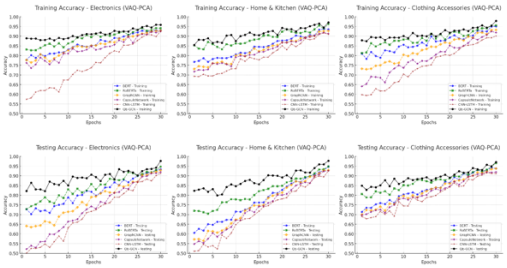


Fig. 5: Evaluation of VAQ-PCA Based Learning Performance

QE-GCN continues to lead under conventional deep learning (ANN-1), with an accuracy of 95–97%. However, when compared to quantum-inspired methods, baseline models perform noticeably worse. GraphCNN stabilizes at a moderate accuracy, but RoBERTa performs better than BERT. QE-GCN outperforms traditional deep learning models in terms of generalization, maintaining testing accuracy between 94 and 96%. CNN-LSTM and CapsuleNetwork still show a lot of volatility and instability.

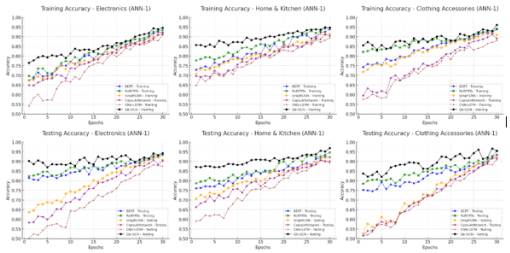


Fig. 6: Evaluation of ANN-1 Based Learning Performance

With a little more sophisticated design, the ANN-2-based graphs display trends that are comparable to those of the ANN-1. While baseline models are still unable to achieve the same training levels, QE-GCN achieves 96–98% accuracy. GraphCNN stabilizes at about 90% accuracy, and RoBERTa outperforms BERT. Across all datasets, QE-GCN continues to be the best-performing model, retaining a high generalization accuracy of approximately 95%. Although baseline models outperform CNN-LSTM and CapsuleNetwork, they are still inferior to the structured learning framework of QE-GCN.

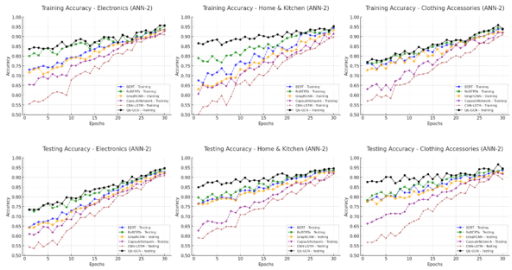


Fig. 7: Evaluation of ANN-2 Based Learning Performance

V. CONCLUSION AND FUTURE SCOPE

Using deep learning contextual feature embeddings and quantum circuit encoding approaches, we presented and thoroughly assessed a quantum-inspired method for sentiment classification in this work. The suggested QE-GCN combines traditional BERT-based contextual embeddings with quantum-inspired techniques including amplitude encoding, entangled feature mappings, and circuit optimization via SPSA. Additionally, to capture subtle sentiment differences among real-world product reviews, QTAE and VAQ-PCA were used in conjunction with smart feature map selections (SQ and 2Q). According to experimental assessments, our quantum-enhanced model achieves higher sentiment classification accuracy and performs noticeably better than deep learning baselines like BERT, RoBERTa, GraphCNN, Capsule Networks, and CNN-LSTM.A focused case study further shown how QE-GCN outperforms classical methods in capturing

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subtle sentiment intensities, guaranteeing strong generalization across various linguistic complexity. Even though QE-GCN has shown notable advancements in sentiment analysis, there are a number of potential avenues for further research. First, quantum feature representations might be further improved by integrating Variational Quantum Algorithms (VQAs), which could increase the efficiency and interpretability of the model. Secondly, verifying the model's functionality on actual quantum hardware would offer more profound understanding of the usefulness of quantum computing in natural language processing. Further confirming this framework's resilience across many datasets may involve extending it to multilingual or cross-domain sentiment classification tasks. These developments could open up new avenues for quantum-enhanced natural language processing applications, bridging the gap between deep learning and quantum computing for practical text analysis.

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